Soyeb Pervez Jim

Software Engineer

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Professional Summary

Software engineer building production LLM agents, semantic retrieval, and multi-tenant SaaS for e-commerce. Main engineer on Apa, the AI shopping agent in Fuego's native Shopify apps. Maintainer of HyprFM, a C++/Qt file manager with 344 GitHub stars and outside contributors. Co-author of a published Elsevier paper on physics-informed deep learning, with a second under review.

Technical Skills

Languages: TypeScript, JavaScript, Python, C++ (Qt 6/QML), SQL, Bash
AI/ML: LLM agents, tool calling, RAG, vector search, embeddings, AI SDK, PyTorch, ONNX, scikit-learn
Backend: Node.js, Express, Hono, Next.js, React, Postgres, MySQL (Vitess), Redis, Stripe, Shopify APIs
Infrastructure: Docker, GitHub Actions, Hetzner, Dokploy, Cloudflare R2, Vercel, Playwright, Nix

Professional Experience

Software Engineer **June 2025 – Present**
Fuego.io *Remote*

- Main engineer on Apa, the tool-calling AI shopping agent inside Fuego's multi-tenant Shopify mobile apps, in production across live storefronts: streamed responses, with product search, recommendations, reviews, and order help returned as structured tool results.
- Built semantic product search over live store catalogs, kept current for every tenant.
- Built the response caching, rate limiting, and failure recovery that keep the agent fast and affordable under production traffic.
- Automated per-brand context generation from each tenant's own site, and built the admin surface: conversation analytics, token metering, and Stripe billing.

Selected Projects

HyprFM *C++, Qt 6, QML, Wayland*

- Keyboard-first file manager with Miller columns, split panes, async transfers, device mounting, and live-reloading TOML themes; QML presentation split from C++ backend services, shipped on AUR, Flatpak, AppImage, and Nix. 344 stars, 14 outside contributors, and an independent LinuxLinks review.

Neural Network X-Ray *Next.js, React, TypeScript, ONNX Runtime*

- Browser visualizer showing activations across all 13 CNN layers as users draw Latin or Bengali characters, inference running locally through WebAssembly. Trained the 146-class model on roughly 980,000 EMNIST and BanglaLekha images, published on Hugging Face.

Research & Publications

Physics-Informed Multi-Task Neural Network for Leak Detection in Multiphase Pipelines *Published, 2026*
Process Safety and Environmental Protection (Elsevier) | doi.org/10.1016/j.psep.2026.109609

- PCE-PINN, a multi-task network predicting leak presence, diameter, and location from inlet/outlet pressure and flow alone: F1 = 1.00 detection and $R^2 = 0.906$ and 0.941 under stratified 5-fold cross-validation, a 16.5% location gain over the best classical baseline.

Physics-Informed Embed-to-Control with Per-Producer MLP Heads for CO_2 Storage Surrogate Modelling
Fuel Processing Technology (Elsevier), submitted 2026 *Under review*

- Replaced the shared linear well model in a reservoir surrogate with per-producer MLP heads trained under physics residuals: saturation RMSE −38%, pressure RMSE −35%, per-well R^2 from 0.82 to 0.91.

Challenge papers. AI4Pain 2026 at ACII (under review): physiological pain classifier, 0.572 test accuracy against a 0.398 baseline. MUMIDC at ABC2026: activity recognition from EEG, EMG, and IMU, 87.9% macro F1.

Education & Achievements

BSc in Electrical and Electronic Engineering, University of Dhaka *2021 – 2026*
Final-year thesis on abnormal activity recognition | EF SET English certificate: C1

Competitive programming. Codeforces peak 1714 (Expert); CodeChef peak 2032 (Div 1). ICPC Dhaka Regional finalist twice: 37th/308 (2024), 37th/225 (2023). National Collegiate Programming Contest 27th/196; Meta Hacker Cup round 2 in 2023 and 2024.

Olympiad. Bangladesh Mathematical Olympiad national first runner-up (2017), second runner-up (2016); co-authored *Ganit Olympiader Hatekhori*, an olympiad primer (2024).